

MATH 4340/ COSC 4340 – Numerical Methods for Differential Equations

HW 10: Due by 11:59pm on Monday April 29, 2024.

Submit to WyoCourses a Jupyter notebook (both ipynb and HTML files) .

Handwritten work should be included as a PDF or typed into the notebook as LaTeX.

1. [20 points] Our notes on the proof to the CFL stability condition (i.e., which requires $ck/h \leq 1$ for the wave equation) had a few “assumed” steps. Fill in the missing steps below.
 - (A) [5 points] State the CFL condition and why it’s important to solving the wave equation using the finite difference method.
 - (B) [5 points] Prove that the eigenvalues λ_i of B satisfy $\lambda_i \in [2 - 4\sigma^2, 2]$. (Hint: See the class notes for slide 12.)
 - (C) [5 points] Prove that if λ is a complex number, then $\lambda + 1/\lambda$ is a real number if and only if $|\lambda| = 1$ or λ is actually a real number.
 - (D) [5 points] Prove that if λ is a real number and that $|\lambda + 1/\lambda| \leq 2$, then $|\lambda| = 1$.
2. [35 points] Consider the Fisher’s equation $u_t = 0.05u_{xx} + g(u)$ with homogeneous Neumann boundary conditions and initial condition $u(x, 0) = 1/2 + \cos(2\pi x)$ with $x \in [0, 1]$ and $t \in [0, 2]$.
 - (A) [5 points] Find all the stable equilibrium points to the PDE when $g(u) = u(u - 1)(2 - u)$.
 - (B) [10 points] Let $h = k = 0.05$, and implement the finite difference method to approximately solve for $u_{i,j}$. Use Python’s root-finding algorithm ‘`scipy.optimize.root`’ for each nonlinear implicit step. Plot your solution as a surface plot.
 - (C) [10 points] Use the definitions of $f_i(\mathbf{z})$ from class to derive the Jacobian matrix J (which is slightly different from J on slide 35 of the class notes). Then implement Newtons method to repeat Problem 2B.
 - (D) [10 points] Repeat Problem 2C but change the initial condition to $f(x) = \frac{3}{2} - \cos(2\pi x)$. How does this change affect the solution as it approaches a stable equilibrium?
3. [45 points] Consider the Burgers equation $u_t + uu_x = 0.05u_{xx}$ with initial condition $u(x, 0) = \frac{2D\beta\pi \sin(\pi x)}{\alpha + \beta \cos(\pi x)}$ for $x \in [0, 1]$ with $\alpha = 5$ and $\beta = 4$ and homogenous Dirichlet boundary conditions $u(0, t) = 0$ and $u(1, t) = 0$ with $t \in [0, 2]$.
 - (A) [10 points] Prove that $u(x, t) = \frac{2D\beta\pi e^{-D\pi^2 t} \sin(\pi x)}{\alpha + \beta e^{-D\pi^2 t} \cos(\pi x)}$ solves the boundary and initial conditions, and make a surface plot of $u(x, t)$ for $x \in [0, 1]$ and $t \in [0, 2]$.
 - (B) [10 points] Let $h = 0.01$ and $k = 0.04$ (i.e., $n = 1/h = 100$ spatial steps and $N = 2/k = 50$ time steps), and implement the finite difference method to approximately solve for $u_{i,j}$. Use ‘`scipy.optimize.root`’ for each nonlinear implicit step. Plot both the approximation $u_{i,j}$ and the approximation error using surface plots for $x \in [0, 1]$ and $t \in [0, 2]$.
 - (C) [10 points] Find the Jacobian J for the nonlinear system, and repeat Problem 3B but use Newton’s root-finding algorithm instead.
 - (D) [15 points] Fix $h = 0.01$ and study how the approximation error $E(x_i, t_j) = |u(x_i, t_j) - u_{i,j}|$ changes as you vary: $k \in \{0.04, 0.02, 0.01\}$. Specifically, repeat Problem 3C for the two other choices for k , and plot the error $E(0.5, 1)$ versus k . (Hint: See the table on page 421 of the book for comparison.)